

BelowTrace

When sewage enters a Detroit basement, turn public records into a truthful, address-specific action record.

RELIABLE TRANSPORTATION & INFRASTRUCTURE, AND SUSTAINABILITY

Built by Reff Wu · A working resident tool, not a map demo.

01

A case file that tells residents what is known, what is unknown, and what to do next.

A Detroit basement backup demands a next action before the evidence exists.

First users are households facing a new backup or private sewer concern. They need to report, document, and find the right official route before hiring repair work.

14, 115

Detroit 311 water-in-basement requests since January 2023

Citywide snapshot dated Sept. 18, 2026. These are service requests, not confirmed incidents or unique households.

THE GAP

Public records, DWSD guidance, program rules, and claim steps sit on separate pages. A resident still has to decide what to do first.

THE RISK

A nearby sewer record cannot prove a private connection. Only a camera inspection can establish the private line's condition.

THE USER

A person dealing with water in a basement needs a truthful record before calling, cleaning, spending money, or applying.

Three steps from an address to a usable action record.

1 · LOCATE

Read address-specific public records, nearby reports, program boundaries, and source dates.

2 · RECORD

Keep the resident's observed facts, service-request number, evidence, and deadlines. Label unknowns.

3 · HAND OFF

Print one case record for DWSD, a plumber, an insurer, or a repair program.

The workflow repeats for each address and never invents a private pipe, a claim outcome, or a City decision.

One end-to-end task: report, record, hand off.

Live at belowtrace.vercel.app. Use 16776 Prevost St as the prepared demo address.

01 · START

Enter an address. BelowTrace returns only recorded public data, program boundaries, and explicit unknowns.

02 · RECORD

Call DWSD. Save the service-request number and observed facts. An attempted call cannot advance the case.

03 · HAND OFF

Print the address-specific case file with dates, evidence, next verified question, and official contacts.

LIVE DEMO: choose “I tried, but did not reach them” and show the case remains on step 1. Then start a separate address to show cases never mix.

Detroit impact, if adopted

BelowTrace serves the City framework priority: Reliable Transportation & Infrastructure, and Sustainability.

5,053

water-in-basement 311 requests since 2023 came from outside PSRP neighborhoods

These are requests, not unique households. The count identifies where an address-specific guide may help before a program route is known.

FIRST DETROIT FOCUS

Households in places such as Fitzgerald/Marygrove, Nolan, and Brightmoor that reported a basement-water concern outside a PSRP neighborhood.

MATERIAL CHANGE

One action record tells a resident what the City data says, what it cannot say, and which question to ask next before spending money.

BEYOND DETROIT

The workflow can travel to cities with public infrastructure records and repair programs. Each city's rules and sources must be mapped locally.

First users, a credible path, and one clear ask.

The first market is specific: Detroit households navigating a new basement-water report or a private sewer concern.

FIRST USERS

Households who just filed or need to file a water-in-basement report, especially where a repair-program route is unclear.

REACH + SIZE

Start through one neighborhood or housing-service partner. The observed Detroit signal is 14,115 relevant 311 requests since 2023, not 14,115 customers.

SUSTAINABILITY

Keep the resident tool free. After pilot evidence, offer paid program-navigation pilots and aggregate, privacy-safe service insights to nonprofit or municipal partners.

ASK: one Detroit neighborhood or housing-service partner, one sewer or housing expert, and 8–12 voluntary residents for a consent-based pilot.